

Understanding Anemia Analysis

Tyler Johnson

Williamson College of Business Administration

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Anemia is a common blood disorder that is known worldwide for its extreme impact on patient's health. Anemia can affect those of all different type of all diverse types of life due to its global effect. This condition is when blood does not contain enough healthy red blood cells or hemoglobin to transport oxygen through the body. Some common symptoms of Anemia fatigue, lack of cognitive function, reduced quality of life, weakness, and in some cases death. Many forms of anemia currently exist to include; iron deficiency, Macrocytic, normocytic, leukemia, Leukemia with thrombocytopenia, and other types of anemia disorders affecting the world. Each of which differ in causation, severity, and treatment for these various forms of anemia. Accurate treatment and classification of these anemia factors are very important as to intentionally increase quality of life and mitigate death due to this blood disease.

An extremely influential factor in dealing with these blood diseases is relying on the laboratory results to evaluate patients for anemia classification or other blood disorders. Measures such as hemoglobin (HGB), red blood cell count (RBC), and mean corpuscular volume (MCV) provide statistically significant results regarding size, concentration, and quality of the blood cells. These lab results are used to help determine between healthy individuals and which blood disorder one may have. Interpreting many various blood results can be confusing and difficult to understand especially when trying to determine which anemia is present in the patient. Utilizing analytical methods to on==uncover trends in the data can be highly effective in creating efficient and proactive patient care allowing more patients to receive higher quality of care and reducing strain on the healthcare industry.

Healthcare analytics is an extremely important field that has grown rapidly in need of these statistical analysis techniques to alleviate a strained and increasingly more demanding field. This type of analytics aims to apply statistical techniques and reasoning, predictive

modeling, and data-based decision making to help solve healthcare data problems. Modeling can identify relationships between the patient demographics and the clinical outcomes to assist these professionals in seeing these patterns and being able to develop an action plan based on these results. Pattern recognition may not immediately be present when analyzing these results without the aid of these models. Commonly in the healthcare analytics field logistical regressions are built to advance research and influence how predictor variables influence outcomes of a result. This specific set of data contains more than two categories; therefore, multinomial logistic regression provides a clear and appropriate data structure for each outcome category or diagnosis.

This study is to examine lab vectors to predict the impact to the diagnosis of anemia classification. This dataset aims to investigate the relationship between hemoglobin, Mean corpuscular volume, and red blood cell count. The model chosen was developed because of the multiple diagnosis values used to determine the impact to these other variables. The findings in this analysis aims to provide insights on the blood results and their relation to anemia blood diseases and how this information can support decision making.

Methods

This study utilized data from a publicly available resource known as Kaggle. It contains a classification variable Diagnosis and various other blood lab results. This dataset was ultimately selected for developing a statistical inference on due to the ability to analyze blood results and compare to classification variable. It also included many variables that are typically used in healthcare for making predictions on. No time component was included in this dataset as it only patient lab results.

The dependent variable in the study was diagnosis. This was a categorical variable that contains multiple diagnosis variables as well as a healthy reference category. The model chosen was the multinomial logistic regression model because the diagnosis variable contains more than two categories. This would be the best model to analyze the results and further research studies relating to anemia.

Three hematological vectors were selected as independent variables Hemoglobin, Mean Corpuscular Volume (MCV), and Red Blood Cell Count (RBC).

Hemoglobin (HGB) – This helps to transport oxygen in the body from the lungs to tissue and organs. Without this oxygen transport organs and tissue cannot properly function creating issues and leading to the adverse effects.

Mean Corpuscular Volume – This is a blood test that measures the average size of the red blood cells. Inconsistencies here can be a huge clue into what the issue may be. This result is typically obtained by doing a complete blood cell count.

Red Blood Cell Count – this blood test is measuring the number of red blood cells are in a sample of blood.

The primary explanatory variable was chosen to be Hemoglobin this is one of the most critical indicators of anemia. Many anemia types can be determined based on the size of the red blood cell therefore mean corpuscular volume was included to account for these instances. Red blood cell count is important because it provides a key detail in the quantity of cells in the blood. The best interpretation of this dataset was to use health responses as the reference for these measures and conduct an analysis. All provided coefficients are being compared to health patients who do not suffer from anemia.

	HGB	MCV	RBC
HGB	1.0000	0.0235	0.4634
MCV	0.0235	1.0000	-0.0396
RBC	0.4634	-0.0396	1.0000

Figure 1: Correlation matrix

Prior to estimating any results for this model. Exploratory analysis was completed to gather an initial assessment of the current data. A correlation matrix was completed to highlight the potential linear relationship between these variables and to assess for multicollinearity because results that rely too heavily on each other would fail to provide a strong analysis of the dependent variable. The dependent variable diagnosis was excluded from this correlation matrix as it was not a continuous variable it is categorical in nature.

This matrix highlights the three main variables used in this analysis and their interaction with one another. The important implication here is that these factors do not have a positive or negative correlation of 0.7. A value that exceeds this margin would have a multicollinearity issue and would be stating that the variables are too heavily reliant on each other to be able to create an accurate model for analysis. This dataset did not reflect this issue so further analysis was conducted to strengthen the understanding of our variables in this dataset.

Statistical Analysis

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> results
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	Diagnosis	Predictor	Coefficient	StdError	zStatistic	pValue
1	Iron deficiency anemia	(Intercept)	103.905601213	0.792859681	131.051690	0.000000e+00
2	Iron deficiency anemia	HGB	13.369182972	3.782463139	3.534518	4.085201e-04
3	Iron deficiency anemia	MCV	9.350988519	0.224270211	41.695188	0.000000e+00
4	Iron deficiency anemia	RBC	21.051219614	2.247518331	9.366428	0.000000e+00
5	Leukemia	(Intercept)	29.320810938	2.179646221	13.452096	0.000000e+00
6	Leukemia	HGB	29.507932266	2.125902504	13.880191	0.000000e+00
7	Leukemia	MCV	95.535291035	0.890716025	107.256733	0.000000e+00
8	Leukemia	RBC	-2.460551833	0.978536125	-2.514523	1.191934e-02
9	Leukemia with thrombocytopenia	(Intercept)	-1.787161204	0.180150685	-9.920369	0.000000e+00
10	Leukemia with thrombocytopenia	HGB	0.630423069	0.162228828	3.886011	1.019047e-04
11	Leukemia with thrombocytopenia	MCV	0.615810003	0.286989840	2.145755	3.189250e-02
12	Leukemia with thrombocytopenia	RBC	-0.318633878	0.255407150	-1.247553	2.121949e-01
13	Macrocytic anemia	(Intercept)	-2.008224387	0.173925865	-11.546439	0.000000e+00
14	Macrocytic anemia	HGB	-0.383398692	0.148113666	-2.588544	9.638273e-03
15	Macrocytic anemia	MCV	-0.438810590	0.178595125	-2.457013	1.400975e-02
16	Macrocytic anemia	RBC	0.844221234	0.133087614	6.343349	2.248228e-10
17	Normocytic hypochromic anemia	(Intercept)	-1.061202597	0.013798113	-76.909255	0.000000e+00
18	Normocytic hypochromic anemia	HGB	-0.161299553	0.038647319	-4.173629	2.997864e-05
19	Normocytic hypochromic anemia	MCV	-0.136293559	0.011594463	-11.755056	0.000000e+00
20	Normocytic hypochromic anemia	RBC	0.005518714	0.004328926	1.274846	2.023637e-01
21	Normocytic normochromic anemia	(Intercept)	-0.089123225	0.024826288	-3.589873	3.308389e-04
22	Normocytic normochromic anemia	HGB	-0.132002061	0.022662763	-5.824624	5.724149e-09
23	Normocytic normochromic anemia	MCV	-1.071357507	0.013695224	-78.228548	0.000000e+00
24	Normocytic normochromic anemia	RBC	0.002957964	0.002745033	1.077570	2.812259e-01
25	Other microcytic anemia	(Intercept)	0.805554077	0.348793664	2.309543	2.091345e-02
26	Other microcytic anemia	HGB	-1.929669297	0.441617089	-4.369553	1.245011e-05
27	Other microcytic anemia	MCV	-1.814437213	0.816771893	-2.221474	2.631890e-02
28	Other microcytic anemia	RBC	-4.878303968	0.717632458	-6.797775	1.062461e-11
29	Thrombocytopenia	(Intercept)	0.782293737	0.348110320	2.247258	2.462354e-02
30	Thrombocytopenia	HGB	-2.794509172	0.331712880	-8.424482	0.000000e+00
31	Thrombocytopenia	MCV	-0.643773092	0.380143132	-1.693502	9.035999e-02
32	Thrombocytopenia	RBC	-2.208097379	0.334638417	-6.598458	4.154566e-11

Figure 2: Multinomial Logistic Regression

The completed regression coefficients highlight an interesting relationship in this model. Many P values for the diagnosis variables are noted as $P < 0.01$. This is as expected since these variables are all significant to anemia analysis. It is also noted that MCV and RBC does have more statistically insignificant values than hemoglobin does. Hemoglobin is the most statistically significant variable when analyzing these anemia blood results.

Discussion

This study was meant to analyze if hematological lab results could be used to predict anemia type. The results indicated that these vectors do provide very meaningful information in predicting the type of anemia one is likely to get or have. Creating this multinomial logistic regression model provided this analysis to conclude this strong relationship for predicting anemia. This model is structured to aid in anemia research and explore the relationship between numerous factors. It also helps to provide strong reasoning behind evidence-based decision

making and be utilized to conduct further analysis and project different outcomes for diverse groups of patients.

Several factors should be considered due to this data being strictly blood results and diagnosis and not including socioeconomic or time series information. Due to this lack of information this dataset cannot study the effect these results may have on people of different race, gender, occupation, or income status. To further advance this research it would be ideal to have important vectors that include that information to look at how patients are being affected. Being able to analyze socioeconomic vectors as such would help to create an understanding if a certain group is facing higher risk compared to other groups. This data lacking a time series also makes this information challenging to get an understanding of when these events were happening. Socioeconomic and time-based data are very important to understanding who and why this issue may be happening.

These limitations did not interfere with the overall goal of this study. This study still demonstrated that blood results are effective predictors in anemia diagnosis and supports clinically driven research. This study also highlights the impact that analysis has on healthcare analytics. The results still suggested the important data driven decision making that can be completed with data analysis of blood results.

Hemoglobin did however prove to be one of the most significant predictors across these many anemia classifications. This is consistent with clinical findings as it is one of the driving factors that typically is used for indication of an anemia blood issue. Reduced hemoglobin levels typically are indication of reduced oxygen flowing through the blood stream and often the leading variable in identifying anemia. Hemoglobin did prove to be very statistically significant

in many diagnosis variables. This would strongly support that changes in blood oxygen levels does strongly support anemia diagnosis heavily.

Another statistically significant variable tested was the Mean Corpuscular Volume. This variable is the average size of the blood cells so fluctuations here would also strongly indicate an issue. Which remained consistent in the findings. This variable can be used to help differentiate between different anemia subtypes. Microcytic, Normocytic, and Macrocytic anemias are all often differentiated by the cell size. This study shows that this is a consistent and important variable for anemia diagnosis. This is another strong variable for clinical assessment of anemia types.

Red Blood Cell Count was a statistically significant variable for some anemia diagnosis but not for all the anemia diagnosis types. This result was interesting to discover as it suggests that red blood cell count is important for some but not for others. This is useful context because it highlights just how variables are interacting with one another to lead to a proper diagnosis. This variation helps to understand that there are fluctuations in what results are and are not useful. Further analysis with more variables would reveal patterns that can be confidently trusted in to create efficient and easy to access healthcare.

Overall, this data suggests that blood laboratory results can be used to create data effective models and can be reliable for classifying anemia types. The multinomial logistic regression model was able to identify statistically significant and insignificant relationships between these different measures. This highlights the predictive capabilities of healthcare analytics as further research studies and technologies are conducted. This model would be effective in supporting decision making for healthcare professionals.

Conclusion

Anemia is one of the most common blood disorders worldwide and has many difficulties for diagnosis and treatment options. Creating efficiencies in predicting which anemia type a patient has will generate more effective treatment plans as well as create more time for treatment. If a patient comes in for routine health checks it is important to notice signs of an issue coming long before they are present to get the best and most impactful care possible. Having accurate classification is very important because different anemia types require different treatments and various levels of care. These analytical models such as the one conducted in this research study provide healthcare organizations the opportunity to maximize the treatment time needed for patients who have this related blood issue as well as support high-quality decision-making techniques to care for more individuals at once.

This study was designed to investigate blood results ability to predict anemia classification. A multinomial logistic regression model was developed based on the variable diagnosis to generate statistical conclusions and support decision making processes via statistical research. The predictor variables hemoglobin, mean corpuscular volume, and red blood cell count were used to generate these results. To get these statistics the data was compared to health individuals with no reported anemia diagnosis to interpret the model.

The results concluded that these variables were valuable predictors for anemia classification. HGB and MCV were highlighted as statistically significant predictors with RBC being useful but not as much for the outcome of this study. These results remained consistent with currently established clinical blood analysis research and helps to drive this research by concluding the effective reasoning behind this information.

This study helps to show the statistical reasoning as to why blood results can be utilized in data analysis to generate meaningful relationships used by healthcare professionals. Having statistical reasoning to generate repeatable and logical results creates an improved healthcare system that allows for a deeper understanding as to how things work and are improving patients' quality of life and generating more treated patients. Data driven approaches to healthcare analytics strengthen diagnostic accuracy, create an efficient and standardized results, and improve patient care.

This dataset contained a limited dataset utilizing a small, selected set of predictors and categorical variables. These results still generated useful and meaningful analysis. The dataset was able to perfectly highlight this regression models capabilities and predictive performance. Further research would include adding more clinical variables, demographics, and socioeconomic variables to further improve predictive performance and generate further insights for analysis. This analysis is showing the true capabilities of statistical analysis as it relates to a healthcare field. Successful transformation of blood data into an actionable machine learning model was successfully completed and furthers the understanding of healthcare analytics.

Reference

Aboelnaga, E. (n.d.) Anemia Types Classification [Data set]. Kaggle

<https://www.kaggle.com/datasets/ehababoelnaga/anemia-types-classification/data>